E–STOP /GUARD

Assessed channels
Devices selected later

→

SAFETY DEVICE

Independent controller
Manual reset /EDM

→

SAFE ACTUATION

Assessed contactors
or drive safe function

→

HAZARD STOP

Remove hazardous energy
Verify stopping behavior

RESET AND FEEDBACK

Safety reset is a separate deliberate operator action.
Feedback/EDM monitors the selected final elements.
No ordinary PLC output can bypass protective stop.
Final terminals/channels depend on the selected safety product.
Stop category /PLr /architecture must follow risk assessment.

PLC OBSERVATION ONLY

SR–01 non–safety auxiliary PLC normal input.
Guard diagnostic feedback PLC door signal.
PLC closes process commands on loss of feedback.
Monitoring input is not itself the protective safety output.
No Cat/PL/SIL conformity is declared for the simulation.

SAFETY FUNCTION CORRESPONDENCE	
Current functional model	Future physical responsibility
M02000 / M02001	Physical E-stop /guard device
M02002 / M02023	Selected safety device manual reset /permission
M01000 / M01007	Auxiliary /diagnostic input to normal PLC
M00202 / M00200	Process RESET and separate START remain distinct

Final safety–device and contactor tags are TBD. Select channels and feedback from the assessed architecture; see drawings 018 and 020.

This sheet is an architecture concept, not a terminal–level safety installation drawing. Use the product manual and perform validation before physical release.

NO.	DATE	REVISION
R0	2026–10–22	CONCEPT / NOT FOR CONSTRUCTION

DWG TITLE

SEMICON SORTER PROJECT
Final Test and Multi–Bin Sorting Cell

PLANNED HARDWARE SAFETY ARCHITECTURE

DRAWN BY	DATE
J. H. KO	2026–10–14

DWG NO
019

SHEET NO
1 OF 1